

Fine-tuning process step by step

1. Navigate to Training folder and prepare the input files for the jsonl generation

Example: createLeadPage

===HTML===

```
<input type="text" class="inputBox" name="companyName" size="30" maxlength="100"
id="createLeadForm_companyName">
```

```
<input type="text" class="inputBox" name="firstName" size="30" maxlength="100"
id="createLeadForm_firstName">
```

```
<input type="text" class="inputBox" name="lastName" size="30" maxlength="100"
id="createLeadForm_lastName">
```

```
<input type="submit" class="smallSubmit" name="submitButton" value="Create Lead" id="ext-
gen635">
```

===JAVA===

```
package com.leaf taps.pages;
```

```
import com.framework.selenium.api.design.Locators;
```

```
import com.framework.testng.api.base.ProjectSpecificMethods;
```

```
/**
```

```
 * Page object representing the Edit Lead page.
```

```
 * Provides methods to update lead details and submit changes.
```

```
 * @author Auto-generated
```

```
 */
```

```
public class CreateLeadPage extends ProjectSpecificMethods {
```

```
    public CreateLeadPage companyName(String companyName) {
```

```
        clearAndType(locateElement(Locators.ID,"createLeadForm_companyName"),
companyName);
```

```
        reportStep(companyName+"Company name entered successfully ", "pass");
```

```
        return this;
```

```

    }

    public CreateLeadPage firstName(String firstName) {
        clearAndType(locateElement(Locators.ID,"createLeadForm_firstName"), firstName);
        reportStep(firstName + "First name entered successfully ", "pass");
        return this;
    }

    public CreateLeadPage lastName(String lastName) {
        clearAndType(locateElement(Locators.ID,"createLeadForm_lastName"), lastName);
        reportStep(lastName + "Last name entered successfully ", "pass");
        return this;
    }

    public CreateLeadPage clickUpdateSubmit() {
        click(locateElement(Locators.CLASS_NAME,"smallSubmit"));
        return new ViewLeadPage();
    }

}

```

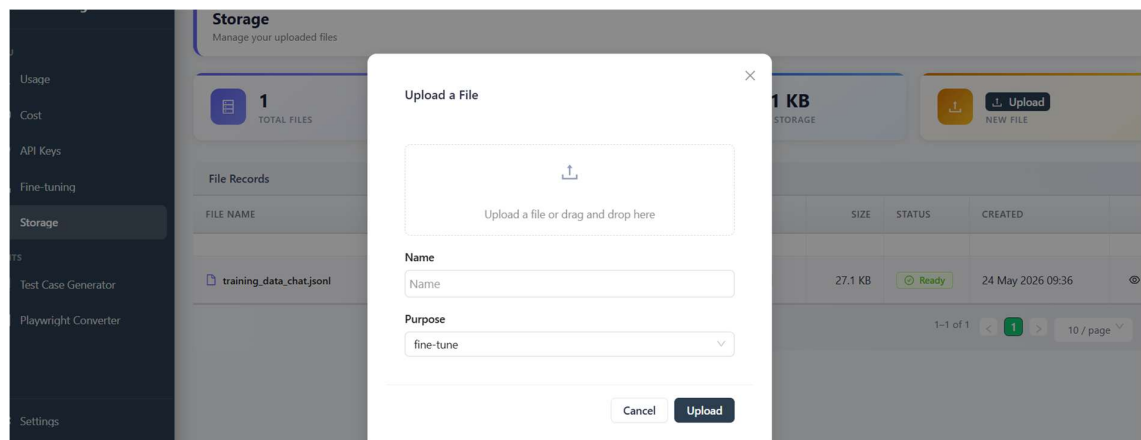
2. Now go to VS code, select Open folder, navigate to leafapps folder path and open it.
3. Navigate to
 \leafapps\src\main\java\com\framework\converter\TrainingDataConverter.java
4. Edit the input and output paths under the main functions

```

42
43
44 public static JSONObject createJsonLine(String html, String java) {
45     String systemMessage = "You are an expert in generating Selenium Java page object clas
46         \"Follow the framework's guidelines: use SeleniumBase wrapper methods, proper l
47
48     String userMessage = \"Given the following DOM structure:\\n```html\\n\" + html + \"\\n```\\n
49         \"Generate a Selenium Java page object class for this DOM.\";
50
51     JSONObject jsonObj = new JSONObject();
52     JSONArray messages = new JSONArray();
53
54     messages.put(new JSONObject().put(\"role\", \"system\").put(\"content\", systemMessage));
55     messages.put(new JSONObject().put(\"role\", \"user\").put(\"content\", userMessage));
56     messages.put(new JSONObject().put(\"role\", \"assistant\").put(\"content\", java));
57
58     jsonObj.put(\"messages\", messages);
59
60     return jsonObj;
61 }
62
63 Run | Debug
64 public static void main(String[] args) {
65     // Adjusted paths to match your specific locations
66     String inputDir = \"E:\\\\Materials\\\\GenAI\\\\Videos\\\\phase-1\\\\week4\\\\day1\\\\training\\\\train
67     String outputFile = \"E:\\\\Materials\\\\GenAI\\\\Videos\\\\phase-1\\\\week4\\\\day1\\\\training\\\\tra
68
69     try (DirectoryStream<Path> stream = Files.newDirectoryStream(Paths.get(inputDir), glob
70         BufferedWriter writer = Files.newBufferedWriter(Paths.get(outputFile),
71             StandardCharsets.UTF_8, StandardOpenOption.CREATE, StandardOpenOption.APP
72

```

5. Now run the main method and the Jsonl file will be generated in the mentioned output folder
6. Once the Jsonl file gets generated, login to <https://api.testleaf.com/>, navigate to storage
7. Click on upload and select the jsonl file, give a valid name and select the purpose as fine-tune and click on upload button.



File Records						
FILE NAME	FILE ID	PURPOSE	SIZE	STATUS	CREATED	
File Name						
training_data_chat.jsonl	file-Bja3X1MZqpexhTJECR6FL5	fine-tune	27.1 KB	Ready	24 May 2026 09:36	

1-1 of 1 < 1 > 10 / page

8. Now navigate to Fine-tuning tab, select the base model, Training data as select existing and select the file which we've uploaded in storage

Batch size as 1, Learning rate multiplier as 0.5 to 0.6 and Number of Epochs as 4-5 and click on create model, It'll take around 20-30mins to complete.

Create a fine-tuned model

METHOD

Supervised

BASE MODEL

gpt-4o-mini-2024-07-18

TRAINING DATA

☐ Upload new ☒ Select existing

file-Bja3X1MZqpexhTJECR6FL5

VALIDATION DATA

☐ Upload new ☐ Select existing

☒ None

SEED

Controls reproducibility. Same seed + parameters produces the same result. Leave 0 to auto-generate.

0

HYPERPARAMETERS

☒ Batch Size

1

☒ Learning Rate Multiplier

0.60

Recommended range: 0.0001 – 10

☒ Number of Epochs

4

Recommended range: 1 – 10

Cancel

Create Model

ft:gpt-4o-mini-2024-07-18:testleaf-2::Dj58d6Fh

✓ Status

🔍 Job ID

🔍 Training Method

📦 Base Model

📦 Output Model

🕒 Created At

Succeeded

ftjob-iOTMVkahFJB9FZDIJf912VDK

supervised

gpt-4o-mini-2024-07-18

ft:gpt-4o-mini-2024-07-18:testleaf-2::Dj58d6Fh

24 May 2026 20:59:15

📦 Trained Tokens

🕒 Epochs

📄 Batch Size

📊 LR Multiplier

🔑 Seed

23,200

4

1

0.6

1521520382

Re-run as Job

Close

9. Once fine-tuning completed navigate to API Keys, click on create keys, give a valid name and click on create, it'll generate the API key, copy it and store it.

Create API Key

* Enter a display name for the key (max 50 characters)

API Key Name

Create Key

API Keys

Create and manage your API keys

0 TOTAL KEYS

API Key Records

NAME

SECRET KEY

USAGE

ACTIONS

+ Create Key

NEW API KEY

API Key Created

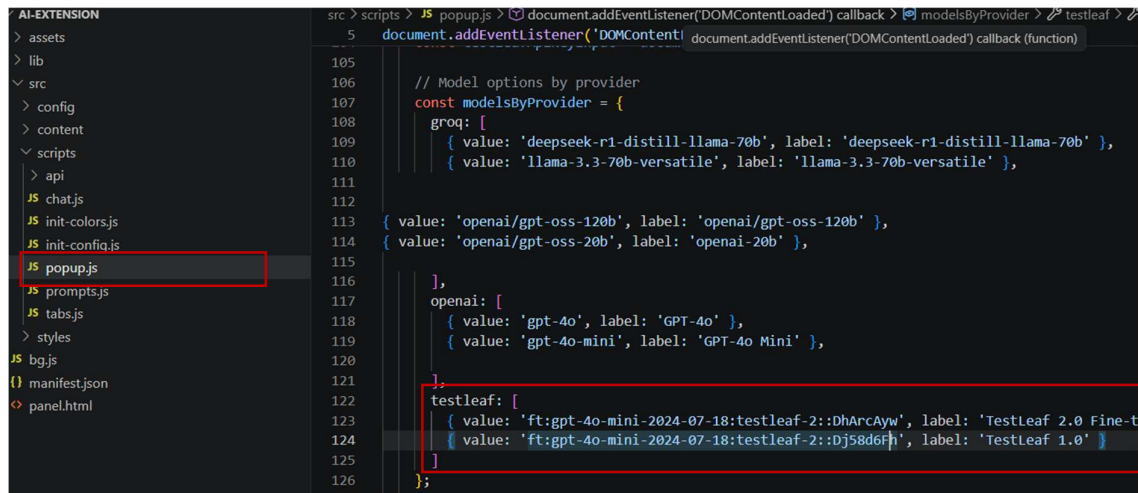
Secret Key:

yW7p84mnuOpq

* This key will only be shown once. Please copy and store it securely.

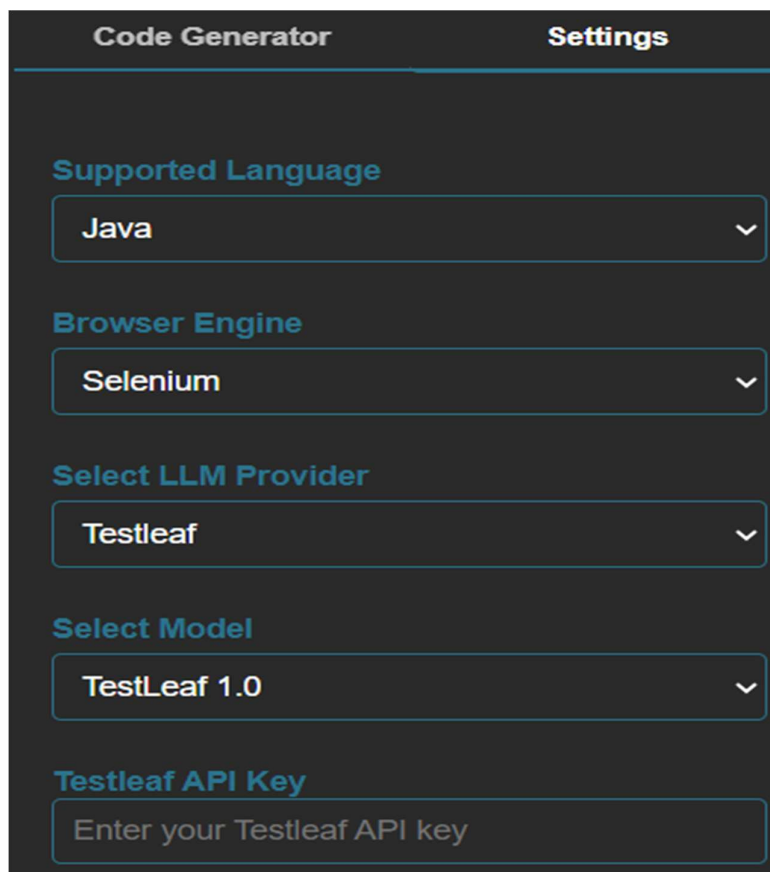
Done

10. Once done, Open the AI extension in VS code, navigate to prompt.js and add the fine-tuned model



```
src > scripts > JS popup.js > document.addEventListener('DOMContentLoaded') callback > modelsByProvider > testleaf >
5 document.addEventListener('DOMContentLoaded') callback (function)
105
106 // Model options by provider
107 const modelsByProvider = {
108   groq: [
109     { value: 'deepseek-r1-distill-llama-70b', label: 'deepseek-r1-distill-llama-70b' },
110     { value: 'llama-3.3-70b-versatile', label: 'llama-3.3-70b-versatile' },
111   ],
112   openai: [
113     { value: 'openai/gpt-oss-120b', label: 'openai/gpt-oss-120b' },
114     { value: 'openai/gpt-oss-20b', label: 'openai-20b' },
115   ],
116   testleaf: [
117     { value: 'ft:gpt-4o-mini-2024-07-18:testleaf-2::DhArcAww', label: 'TestLeaf 2.0 Fine-t' },
118     { value: 'ft:gpt-4o-mini-2024-07-18:testleaf-2::Dj58d6fH', label: 'TestLeaf 1.0' },
119   ],
120 };
121
122
123
124
125
126
127
```

11. Once done, Open the AI extension in browser, go to settings and select the LLM provider and the model as our fine-tuned one and enter the generated API key



Code Generator	Settings
Supported Language	
Java	
Browser Engine	
Selenium	
Select LLM Provider	
Testleaf	
Select Model	
TestLeaf 1.0	
Testleaf API Key	
Enter your Testleaf API key	

12. Inspect and generate the scripts using the fine-tuned model

Feature: Login to OpenTaps

Scenario Outline: Successful login with valid credentials

Given I open the login page

When I type "<username>" into the Username field

And I type "<password>" into the Password field

And I click the Login button

Then I should be logged in successfully

Examples:

username	password	
"srinivas"	"srinivas123"	
"kavitha"	"kavitha123"	
"ravi"	"ravi123"	

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import org.testng.Assert;
import org.testng.annotations.AfterClass;
import org.testng.annotations.BeforeClass;
import org.testng.annotations.Test;
```

```
import java.time.Duration;
```

```
public class LoginTest {
    private WebDriver driver;
```

```
private WebDriverWait wait;
```

```
@BeforeClass
```

```
public void setUp() {
```

```
    // Setup ChromeDriver and WebDriverWait
```

```
    driver = new ChromeDriver();
```

```
    wait = new WebDriverWait(driver, Duration.ofSeconds(10));
```

```
    driver.manage().window().maximize();
```

```
}
```

```
@Test
```

```
public void testLoginWithValidCredentials() {
```

```
    // Navigate to the login page
```

```
    driver.get("https://leaftaps.com/opentaps/control/main");
```

```
    // Enter valid username and password
```

```
    driver.findElement(By.id("username")).sendKeys("Demosalesmanager");
```

```
    driver.findElement(By.id("password")).sendKeys("crmsfa");
```

```
    // Click the login button
```

```
    driver.findElement(By.className("decorativeSubmit")).click();
```

```
    // Wait for the welcome message and verify it
```

```
    wait.until(ExpectedConditions.visibilityOfElementLocated(By.className("decorativeSubmit")));
```

```
    String welcomeMessage = driver.findElement(By.className("decorativeSubmit")).getText();
```

```
    Assert.assertEquals(welcomeMessage, "Welcome Demosalesmanager", "Login failed or  
welcome message is incorrect.");
```

```
}
```

```
@Test
```

```
public void testLoginWithInvalidCredentials() {
```

```

// Navigate to the login page
driver.get("https://leaftaps.com/opentaps/control/main");

// Enter invalid username and password
driver.findElement(By.id("username")).sendKeys("InvalidUser");
driver.findElement(By.id("password")).sendKeys("InvalidPass");

// Click the login button
driver.findElement(By.className("decorativeSubmit")).click();

// Wait for the error message and verify it
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("errorDiv")));
String errorMessage = driver.findElement(By.id("errorDiv")).getText();
Assert.assertTrue(errorMessage.contains("Invalid username or password"), "Error message not
displayed as expected.");
}

@AfterClass
public void tearDown() {
    // Cleanup and close the browser
    if (driver != null) {
        driver.quit();
    }
}
}

```